Crop Type and Disease Classification

GitHub link: https://github.com/lyurish00/DS4002-CS3

Context:

The classification of diseases present in crops using machine learning plays a vital role in modern agriculture by enhancing productivity, sustainability, and food security. Accurate crop type identification allows for better land use planning, yield prediction, and resource allocation, while timely detection and classification of plant diseases can prevent widespread crop damage and economic loss. Machine learning algorithms can analyze large volumes of data from satellite imagery, drones, and field sensors to detect patterns that may be invisible to the human eye, enabling early intervention and more efficient farming practices. By automating these processes, machine learning not only reduces the reliance on manual inspection but also supports precision agriculture, helping farmers make informed decisions based on real-time insights.

Motivation:

You are a data scientist that has been hired by the US Department of Agriculture (USDA) to create a machine learning model to classify what disease is present in each image of a crop, or state that the crop is healthy. The USDA has already developed a model that can classify the crop type, but they specifically need help with developing a model to identify diseases in tomatoes. This model has the potential to increase crop production by catching diseased plants before the disease spreads to the entire field.

Deliverable:

The USDA has asked you to produce a model that can, given the image of a tomato, predict if the tomato is healthy, or it has early blight, septoria spot, bacterial spot, late blight, mosaic virus, yellow virus, or mold. You have been given a dataset containing 744 images of tomatoes that have already been split with a 90/10% train/test split. A random forest model to classify the crop type has been provided, and you will utilize the structure of this model to train and test a random forest model to classify the disease present (if any) in tomato images. You will then utilize a grid search to identify ideal hyperparameter choices and improve the model.